

Education

George Mason University , MS in Computer Science <i>Fairfax, VA, USA</i>	May 2025
Guru Gobind Singh Indrapratha University , B.Tech in Information Technology <i>New Delhi, India</i>	Jul 2023

Experience

Software Developer , <i>College of Public Health, GMU</i>	June 2025 – Present
- Building a secure Flask-based app for bruise image data set segregation to assist department-wide research. - Developed frontend, API integrations, and robust backend for managing bruise image data. - Working on the backend of a Bruise Detection Application using YOLO; integrating 3D human body visualization to display patient injuries. - Application intended for use by healthcare professionals and legal proceedings related to domestic violence cases.	
Research Assistant – Generative Deep Learning , <i>George Mason University</i>	May 2025 – Present
- Working on molecular graph generation using MolGAN to design novel, valid molecules. - Integrating Birdie-style pretraining objectives to enhance diversity and synthesizability. - Aiming to generate high-quality molecular candidates suitable for lab-scale development and drug discovery pipelines.	
Software Developer & Research Assistant , <i>Geoweaver – NASA/ESIP</i>	Feb – Oct 2024
- Led 4 interns on ML workflows for geospatial data and co-authored their first publications. - Added support for PostgreSQL, MySQL, and H2 with CLI integration. Implemented an in-app code editor and improved UI responsiveness. - Reduced processing time by 25% in Snow/Ozone Estimation models and ran large-scale workflows on Hopper supercomputer with 20% speed-up. - Introduced Vitest for unit testing, accelerating development feedback loops and reducing CI runtime.	
Data Science Intern , <i>QBurst Technologies</i>	Jul – Aug 2022
- Improved BigQuery query performance by 15%. Conducted training and documentation improving team output. - Assisted in developing a pipeline to migrate company databases from Amazon RedShift to BigQuery with GCP.	
ML Intern , <i>DynPro</i>	Apr – Jun 2022
- Optimized recommendation systems, leading to a 6.5% gain in prediction accuracy. - Delivered regression and clustering results boosting success rate by 20%.	
Software Engineer Intern , <i>Accops Systems</i>	Jan – Mar 2022
Worked on backend logic using Python and C++, reducing bug resolution time by 15%.	
Software Developer , <i>Tech Solutions</i>	Jan 2020 – Jun 2021
- Developed RESTful APIs using Node.js and Express, enabling CRUD operations for a multi-tenant inventory management platform; integrated JWT-based authentication and role-based access control. - Designed and optimized PostgreSQL database schemas, implementing indexes and query optimization to reduce response latency by 30% for high-traffic endpoints. - Automated CI/CD pipelines using GitHub Actions, reducing deployment overhead and enabling faster iteration across dev/staging/production environments.	
Cloud Trainee , <i>Tech Explica</i>	Jul – Aug 2021
- Deployed websites on AWS using EC2 and S3. - Increased deployment efficiency by 20% by implementing CI/CD pipelines with GitHub Actions.	

Publication

- Socioeconomic and Industrial Factors Influencing U.S. Ozone Pollution – NASA ESIP (AGU).
- Snow Water Equivalent Estimation Using ML & Remote Sensing – NASA ESIP.

Projects

Mason Critique – <i>2nd Place</i>	HackFax 2025
- Developed a full-stack app using React (Node.js, TypeScript), Firebase, and Tailwind CSS, enabling GMU students to review courses and professors with real-time feedback. - Engineered back-end with Firestore and Google OAuth, integrating GPA calculators and Chart.js visualizations.	
FinSight – <i>GenAI Finance Advisor (Google Capstone)</i>	Apr – May 2024
- Full-stack finance platform with React, FastAPI, and Vertex AI Gemini. - Used document-grounded RAG and interactive Recharts for insights.	
Diabetic Retinopathy Detection – <i>ML on Retinal Images</i>	Apr – May 2023
Used EfficientNet-B4, ResNet18, and custom CNN achieving QWK: 0.69.	

Skills and Tools

Languages: Python, C, C++, SQL, Java, Bash Libraries/Tools: PyTorch, TensorFlow, Pandas, NumPy, Docker, Kubernetes, GitHub Actions, Terraform, Redmine	Web & Cloud: HTML, CSS, JS, React, Angular, jQuery, AWS, Azure, GCP, ML/DL, Security
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